



Handling Tables

APR, 2026

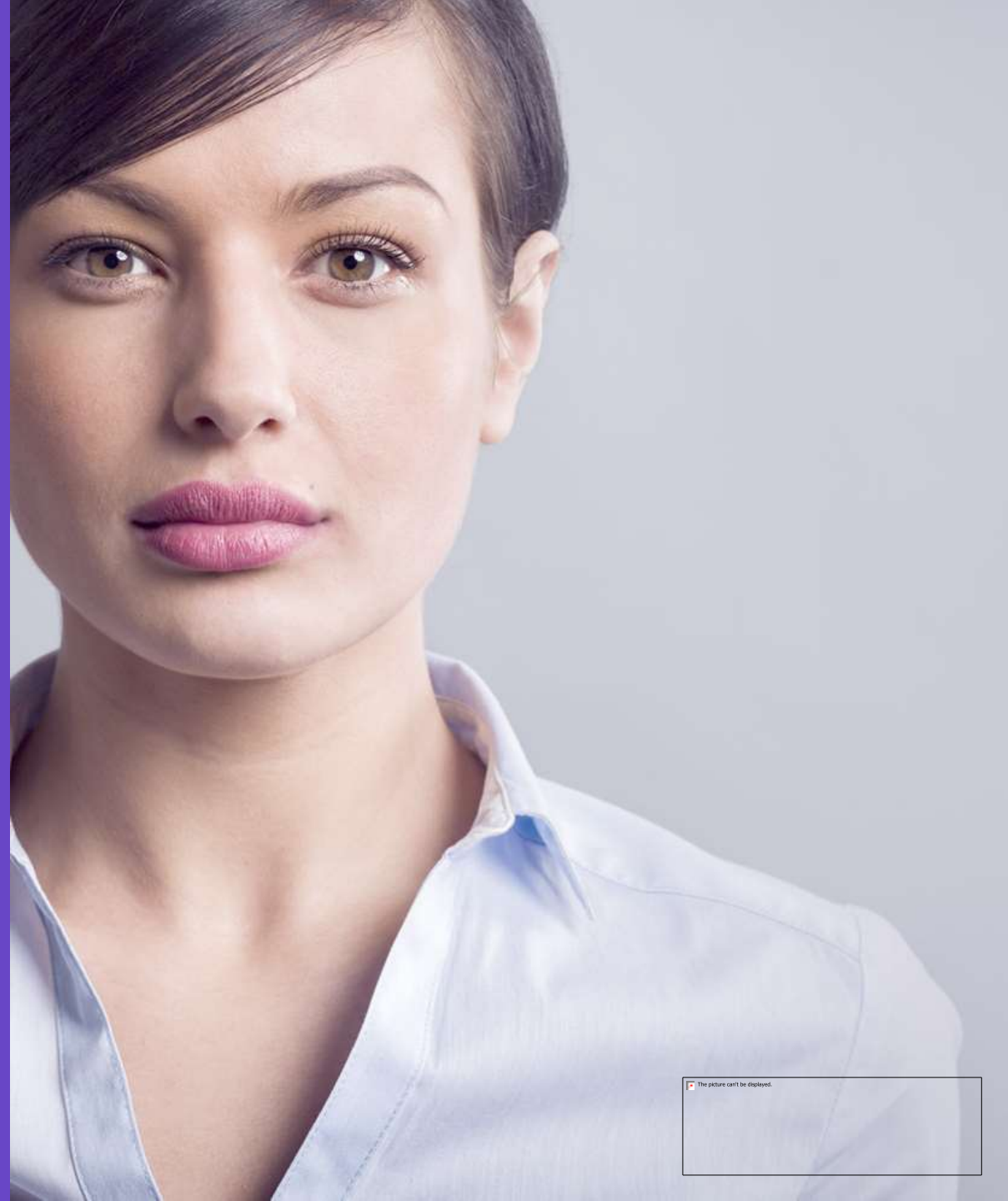
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Contents

- Web Tables and HTML Table Tags
- Types of Web Tables
- Handling Web Tables in Selenium



Web Tables



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Introduction

- **Web table in Selenium is a WebElement** just like any other popular WebElements like text boxes, radio buttons, checkboxes, drop-down menus, etc.
- Web table and its contents can be **accessed using the WebElement functions along with Selenium locators to identify the element (row/column)** on which the operation needs to be performed.
- A table consists of rows and columns. Below are some of the important tags associated with a web table:
 - **< table >** – Defines an HTML table
 - **< th >** – Contains header information in a table
 - **< tr >** – Defines a row in a table
 - **< td >** – Defines a column in a table



Web Tables

Example – Table Creation

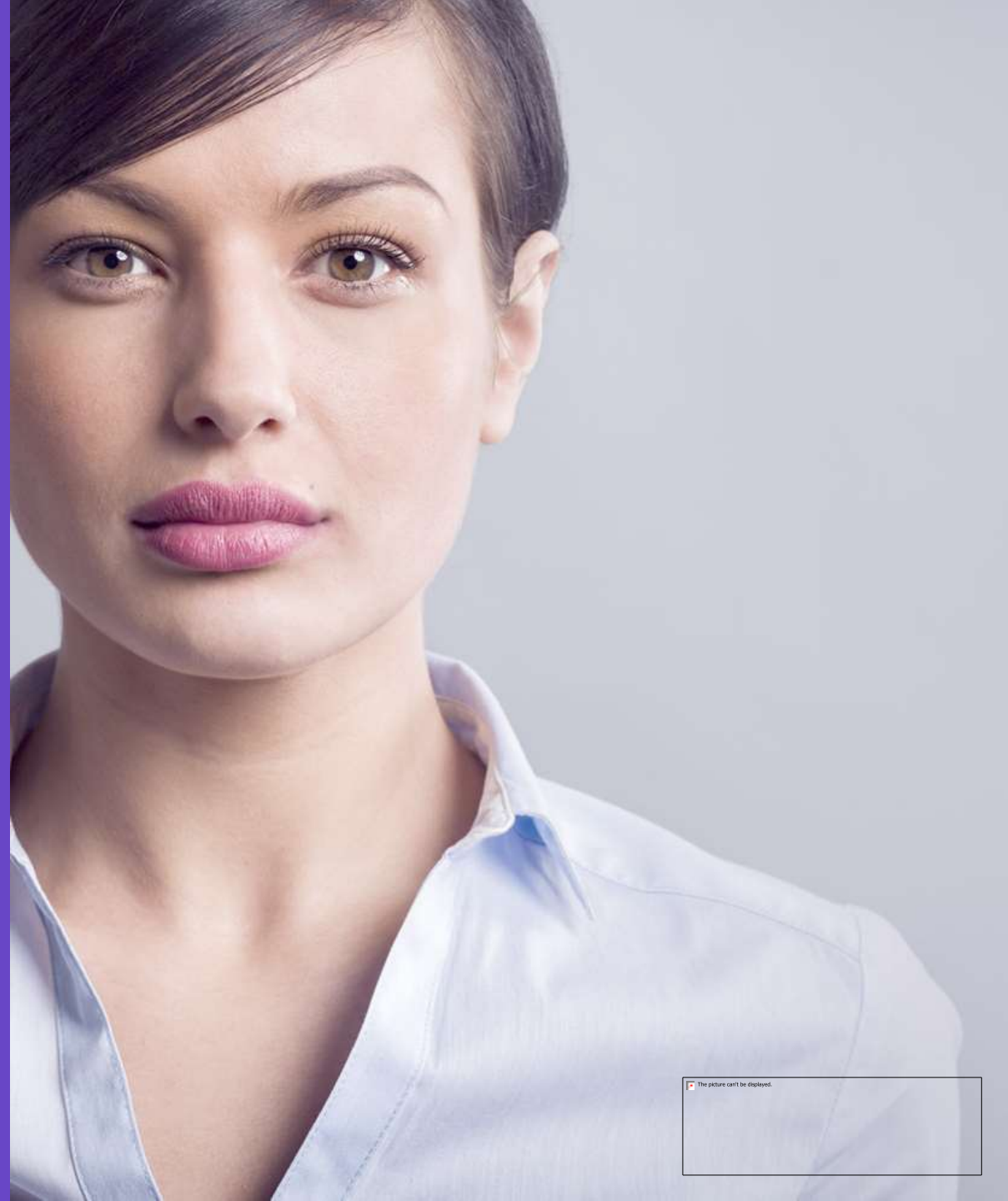
```
<html>
  <table border="5">
    <thead>
      <tr>
        <th>Name</th>
        <th>Age</th>
        <th>Place</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>Arun</td>
        <td>35</td>
        <td>Coimbatore</td>
      </tr>
      <tr>
        <td>Vijay</td>
        <td>37</td>
        <td>Chennai</td>
      </tr>
    </tbody>
  </table>
</html>
```



A screenshot of a web browser window showing the rendered HTML table. The browser's address bar displays the file path: C:/Users/Arun%20Kumar/Downloads/table.html. The table has a thick black border and contains the following data:

Name	Age	Place
Arun	35	Coimbatore
Vijay	37	Chennai

Types of Web Tables



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Types of Web Tables

Web Table Types

- There are two broad categories of tables namely:

1. Static Web Table

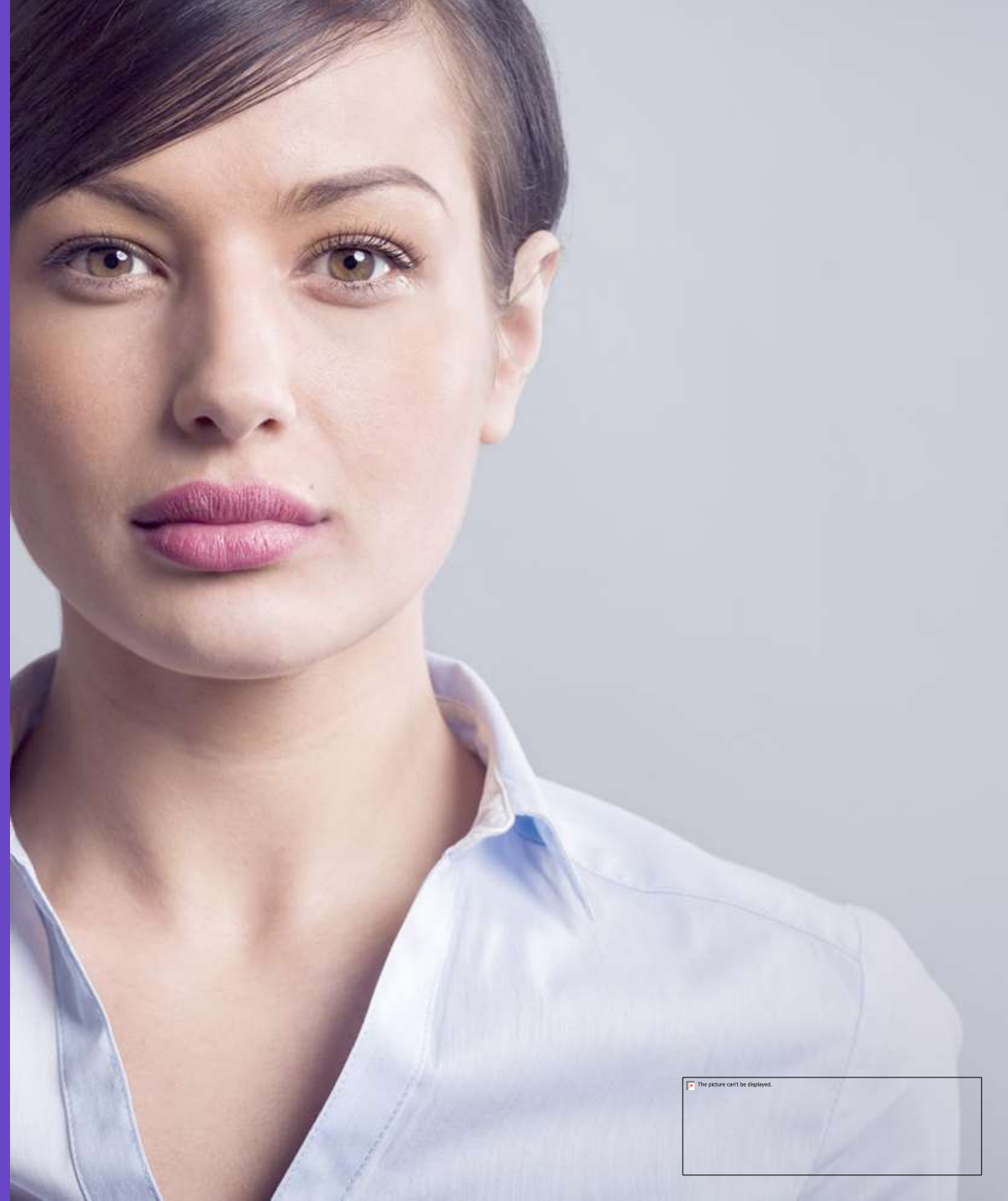
- As the name indicates, the information in the table is static in nature.
- These tables have fixed data that remains unchanged throughout.

2. Dynamic Web Table

- The information displayed in the table is dynamic.
- These tables have data that changes over time, and hence the number of rows and columns might also change depending upon the data shifts.
- E.g. Detailed Product information on e-commerce websites, sales reports, etc.



Handling Web Tables in Selenium



Handling Web Tables in Selenium

Inspecting table and its attributes using XPath

- It is very seldom for a web designer to provide an id or name attribute to a certain cell in the table.
- Therefore, we cannot use the usual methods such as "By.id()", "By.name()", or "By.cssSelector()".
- In this case, the most reliable option is to access them using the "By.xpath()" method.
- Refer the <http://omayo.blogspot.com/> for the testing purpose.



Handling Web Tables in Selenium

Inspecting table and its attributes using XPath

- Xpath is `//table[@id='table1']` - to locate complete table
- Xpath is `//table[@id='table1']/thead` - to locate table heading section
- Xpath is `//table[@id='table1']/tbody` - to locate table body section
- Xpath is `//table[@id='table1']/thead/tr/th`
or
- Xpath is `//table[@id='table1']//th` - to locate all table headings
- Xpath is `//table[@id='table1']/tbody/tr/td`
or
- Xpath is `//table[@id='table1']//td` - to locate all the table data



Handling Web Tables in Selenium

Inspecting table and its attributes using XPath

- Xpath is `//table[@id='table1']/tbody/tr[1]` - to locate first row in the table body
- Xpath is `//table[@id='table1']/tbody/tr[2]` - to locate second row in the table body
- Xpath is `//table[@id='table1']/tbody/tr[3]` - to locate third row in the table body
- Xpath is `//table[@id='table1']/tbody/tr[1]/td` - to locate all the data in the first row
- or
- Xpath is `//table[@id='table1']//tr[1]//td` - to locate all the data in the first row
- Xpath is `//table[@id='table1']//tr[2]//td` - to locate all the data in the second row



Handling Web Tables in Selenium

Inspecting table and its attributes using XPath

- Xpath is `//table[@id='table1']/tbody/tr[1]/td[1]` - to locate the data in 1st row and 1st column
- or
- Xpath is `//table[@id='table1']//tr[1]//td[1]` - to locate the data in 1st row and 1st column
- Xpath is `//table[@id='table1']//tr[2]//td[2]` - to locate the data in 2nd row and 2nd column
- Xpath is `//table[@id='table1']//tr[3]//td[2]` - to locate the data in 3rd row and 2nd column
- Xpath is `//table[@id='table1']//tr//td[3]` - to locate all the data in the 3rd column



Handling Table – Printing Table Headings

```
public class StaticTable {  
public static void main(String[] args)  
{  
    WebDriver driver = new ChromeDriver();  
    driver.manage().window().maximize();  
    driver.get("http://omayo.blogspot.com/");  
  
    List<WebElement> headings = driver.findElements(By.xpath("//table[@id='table1']//th"));  
    System.out.println(headings.size());  
  
    for(WebElement heading : headings)  
    {  
        System.out.println(heading.getText());  
    }  
}  
}
```

Handling Web Tables in Selenium

Output

```
SLF4J: No SLF4J providers were found.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See https://www.slf4j.org/codes.html#noProviders for further details.
Starting ChromeDriver 111.0.5563.64 (c710e93d5b63b7095afe8c2c17df34408078439d-refs/branch-heads/5563@{#995}) on port 4523
Only local connections are allowed.
Please see https://chromedriver.chromium.org/security-considerations for suggestions on keeping ChromeDriver safe.
ChromeDriver was started successfully.
[1683624968.253][WARNING]: This version of ChromeDriver has not been tested with Chrome version 112.
May 09, 2023 3:06:08 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find an exact match for CDP version 112, so returning the closest version found: 111
3
Name
Age
Place
```

Handling Web Tables in Selenium

Handling Table – Printing all Table data

```
public class StaticTable {  
public static void main(String[] args)  
{  
    WebDriver driver = new ChromeDriver();  
    driver.manage().window().maximize();  
    driver.get("http://omayo.blogspot.com/");  
  
    List<WebElement> datas = driver.findElements(By.xpath("//table[@id='table1']/td"));  
    System.out.println(datas.size());  
  
    for(WebElement data : datas)  
    {  
        System.out.println(data.getText());  
    }  
  
}
```

Handling Web Tables in Selenium

Output

```
<terminated> StaticTable [Java Application] C:\Users\Arun Kumar\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_17.0.  
ChromeDriver was started successfully.  
[1683625754.883][WARNING]: This version of ChromeDriver has not been tested with Chrome version 112.  
May 09, 2023 3:19:15 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch  
WARNING: Unable to find an exact match for CDP version 112, so returning the closest version found: 11  
12  
Kishore  
22  
Delhi  
Manish  
25  
Pune  
Praveen  
29  
Bangalore  
Dheepthi  
31  
Mumbai
```


Handling Static Table – To retrieve first row from table

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import java.util.List;

public class SeleniumExample {
    public static void main(String[] args) {

        // Initialize the ChromeDriver
        WebDriver driver = new ChromeDriver();

        // Maximize the window
        driver.manage().window().maximize();

        // Navigate to the URL
        driver.get("https://omayo.blogspot.com/");
```

Handling Static Table – To retrieve first row from table

```
// Retrieve the first row of the table
List<WebElement> tableRow1Data =
driver.findElements(By.xpath("//table[@id='table1']/tbody/tr[1]/td"));

// Print the text of each cell in the first row
for (WebElement data : tableRow1Data) {
    System.out.println(data.getText());
}

// Close the browser
driver.quit();
}
```

```
Kishore
22
Delhi
```

Handling Web Tables in Selenium

Handling Static Table – To retrieve second row third column data from table

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class SeleniumExample {
    public static void main(String[] args) {

        // Initialize the ChromeDriver
        WebDriver driver = new ChromeDriver();

        // Maximize the window
        driver.manage().window().maximize();

        // Navigate to the URL
        driver.get("https://omayo.blogspot.com/");
```

Handling Web Tables in Selenium

Handling Static Table – To retrieve second row third column data from table

```
// Retrieve the second row, third column data from the table
WebElement tableRow2Column3Data =
driver.findElement(By.xpath("//table[@id='table1']//tr[2]/td[3]"));

// Print the text of the element
System.out.println(tableRow2Column3Data.getText());

// Close the browser
driver.quit();
}
```

OUTPUT

Pune

Handling Static Table – To retrieve all data in second column

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import java.util.List;

public class SeleniumExample {
    public static void main(String[] args) {

        // Initialize the ChromeDriver
        WebDriver driver = new ChromeDriver();

        // Maximize the window
        driver.manage().window().maximize();

        // Navigate to the URL
        driver.get("https://omayo.blogspot.com/");
```

Handling Static Table – To retrieve all data in second column

```
// Retrieve all data in the second column of the table
List<WebElement> tableColumn2Data =
driver.findElements(By.xpath("//table[@id='table1']/tbody/tr/td[2]"));

// Print the text of each element in the second column
for (WebElement data : tableColumn2Data) {
    System.out.println(data.getText());
}

// Close the browser
driver.quit();
}
```

```
22
25
29
31
```

Handling Web Tables in Selenium

Handling Static Table – To find number of rows and columns in the table

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import java.util.List;

public class SeleniumExample {
    public static void main(String[] args) {

        // Initialize the ChromeDriver
        WebDriver driver = new ChromeDriver();

        // Maximize the window
        driver.manage().window().maximize();

        // Navigate to the URL
        driver.get("https://omayo.blogspot.com/");
```



Handling Web Tables in Selenium

Handling Static Table – To find number of rows and columns in the table

```
// Find all rows in the table
List<WebElement> rows =
driver.findElements(By.xpath("//table[@id='table1']/tr"));
System.out.println("Number of rows: " + rows.size());

// Find all column headers in the table
List<WebElement> columns =
driver.findElements(By.xpath("//table[@id='table1']/th"));
System.out.println("Number of columns: " + columns.size());

// Close the browser
driver.quit();
}
```

5
3



Handling Static Table – To retrieve entire table

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import java.util.List;

public class SeleniumExample {
    public static void main(String[] args) {

        // Initialize the ChromeDriver
        WebDriver driver = new ChromeDriver();

        // Maximize the window
        driver.manage().window().maximize();

        // Navigate to the URL
        driver.get("https://omayo.blogspot.com/");
```

Handling Static Table – To retrieve entire table

```
// Retrieve the entire table (rows and columns)
List<WebElement> rows =
driver.findElements(By.xpath("//table[@id='table1']//tr"));
int rowCount = rows.size(); // Number of rows

List<WebElement> columns =
driver.findElements(By.xpath("//table[@id='table1']//th"));
int columnsCount = columns.size(); // Number of columns

// Iterate through each row and column to print the data
for (int r = 1; r <= rowCount; r++) {
    for (int c = 1; c <= columnsCount; c++) {
        WebElement data;
        if (r == 1) {
            // For header row, use <th> elements
            data = driver.findElement(By.xpath("//table[@id='table1']//tr[" + r +
                "]/th[" + c + "]]"));
        } else {
```

Handling Web Tables in Selenium

Handling Static Table – To retrieve entire table

```
// For data rows, use <td> elements
    data = driver.findElement(By.xpath("//table[@id='table1']//tr[" + (r) +
    "]/td[" + c + "]"));
    }
    System.out.print(data.getText() + " ");
    }
    System.out.println(); // To move to the next line after printing each row
}

// Close the browser
driver.quit();
}
```

```
5
3
Name Age Place
Kishore 22 Delhi
Manish 25 Pune
Praveen 29 Bangalore
Dheepthi 31 Mumbai
```

Handling Web Tables in Selenium

Handling Dynamic Table – problem in accessing Changing data

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import java.util.concurrent.TimeUnit;

public class SeleniumTest {
    public static void main(String[] args) {
        // Initialize the Chrome WebDriver
        WebDriver driver = new ChromeDriver();

        // Maximize the browser window
        driver.manage().window().maximize();

        // Open the website
        driver.get("https://thinking-tester-contact-list.herokuapp.com/");
```



Handling Dynamic Table – problem in accessing Changing data

```
// Implicit wait for elements to load
driver.manage().timeouts().implicitlyWait(10, TimeUnit.SECONDS);

// Input the email
driver.findElement(By.id("email")).sendKeys("aa.83@gmail.com");

// Input the password
driver.findElement(By.id("password")).sendKeys("Test143$");

// Click on the submit button
driver.findElement(By.id("submit")).click();

// Find the required data in the dynamic table
WebElement requiredData =
driver.findElement(By.xpath("//table[@id='myTable']/tr[2]"));

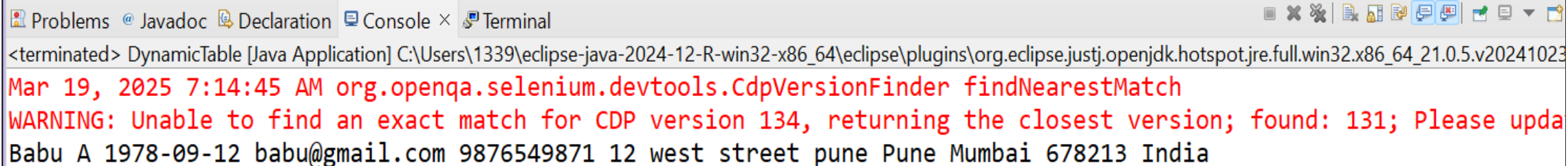
// Print the extracted text
System.out.println(requiredData.getText());

driver.quit(); } }
```

Handling Web Tables in Selenium

Output

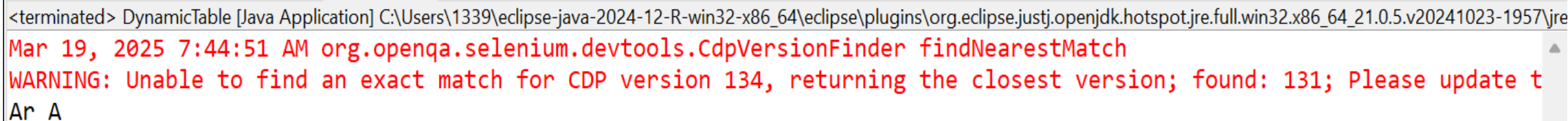
Output: 1st Run



The screenshot shows the Eclipse IDE's console window. The title bar includes 'Problems', 'Javadoc', 'Declaration', 'Console', and 'Terminal'. The console output is as follows:

```
<terminated> DynamicTable [Java Application] C:\Users\1339\eclipse-java-2024-12-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.5.v20241023-1957\jre
Mar 19, 2025 7:14:45 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find an exact match for CDP version 134, returning the closest version; found: 131; Please update to 134
Babu A 1978-09-12 babu@gmail.com 9876549871 12 west street pune Pune Mumbai 678213 India
```

Output: 2nd Run after table got new records



The screenshot shows the Eclipse IDE's console window. The title bar includes 'Problems', 'Javadoc', 'Declaration', 'Console', and 'Terminal'. The console output is as follows:

```
<terminated> DynamicTable [Java Application] C:\Users\1339\eclipse-java-2024-12-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.5.v20241023-1957\jre
Mar 19, 2025 7:44:51 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find an exact match for CDP version 134, returning the closest version; found: 131; Please update to 134
Ar A
```

Handling Web Tables in Selenium

Handling Dynamic Table – Retrieving row data

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import java.util.List;
import java.util.concurrent.TimeUnit;

public class SeleniumTest {
    public static void main(String[] args) {

        // Initialize the Chrome WebDriver
        WebDriver driver = new ChromeDriver();

        // Maximize the browser window
        driver.manage().window().maximize();
        // Open the website
        driver.get("https://thinking-tester-contact-list.herokuapp.com/");
```

Handling Dynamic Table – Retrieving row data

```
// Implicit wait for elements to load
driver.manage().timeouts().implicitlyWait(10, TimeUnit.SECONDS);

// Input the email
driver.findElement(By.id("email")).sendKeys("aa.83@gmail.com");

// Input the password
driver.findElement(By.id("password")).sendKeys("Test143$");

// Click on the submit button
driver.findElement(By.id("submit")).click();

// Define the expected name
String expectedName = "Babu A";
```


Handling Dynamic Table – Retrieving row data

```
// Retrieve contact names from the dynamic table
List<WebElement> contactNames =
driver.findElements(By.xpath("//table[@id='myTable']/tr/td[2]"));

// Get the count of contact names
int contactCount = contactNames.size();
System.out.println("Total contact count: " + contactCount);

// Print all contact names
for (WebElement name : contactNames) {
    System.out.println(name.getText());
}
```

Handling Dynamic Table – Retrieving row data

```
// Loop to find the expected name and print the entire row data
int i = 1;
for (WebElement name : contactNames) {
    if (name.getText().equals(expectedName)) {
        // Retrieve the full row data for the matched name
        List<WebElement> actualRowData =
driver.findElements(By.xpath("//table[@id='myTable']/tr[" + i + "]"));
        for (WebElement row : actualRowData) {
            System.out.println(row.getText());
        }
    }
    i++;
}
driver.quit();
}}
```

Handling Web Tables in Selenium

Output

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code



[Running] python -u "c:\Users\Arun Kumar\VSCodePython\dynamicTable2.py"

4

A A

Ajay A

Babu A

raj k

Babu A 1978-09-12 babu@gmail.com 9876549871 12 west street pune Pune Mumbai 678213 India

[Done] exited with code=0 in 16.395 seconds

